

Energy and the occupant's thermal perception of low-income dwellings in hot-dry climate: Mexicali, México

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ABSTRACT

The residential sector in Mexico requires around 25% of the national electric energy consumption. In hot-dry climate cities like Mexicali, the per capita consumption is 5924 MWh/year, 3.5 times more than the national average. The high temperatures during summer in the city generate conditions of thermal discomfort in dwellings. Therefore, the use of air conditioning equipment and non thermal adequate constructions promote high electrical consumptions. The objective of this study is to determine how electrical consumption in the low-income dwellings depends on the characteristics of design and construction or the occupant's perception has of the conditions of thermal comfort. A field study was done, consisting on the application of surveys. Electrical energy consumption of low-income dwelling was obtained and related with the perception of thermal comfort. The results showed that in this kind of houses, the electricity energy consumption was higher than the average of Mexicali's dwellings, due to the conditions of design and construction of the houses and particularly, by occupant's behavior.

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1. Introduction

Energy efficiency and climate change are topical issues the world over. The building sector plays an important role as it accounts for significant high percentages of national energy consumption. The climate determines significantly the energy consumption, which is why the gases of the greenhouse effect are higher on arid zones, because of the frequent use of the air conditioning.

The residential sector in Mexico requires around 25% of the national electric energy consumption [1] and contributes with approximately 8% of the greenhouse effect gases, including the direct ones, which come from the electricity consumption, and the indirect which come from the use of fossil fuels [2]. In hot-dry climate cities like Mexicali, the per capita consumption is 5924 MWh/year, 3.5 times more than the national average. The high temperatures during summer about 41.8 °C (maximum average temperature) during July and August in the city generate conditions of thermal discomfort in dwellings. Therefore, the use of air conditioning equipment and non thermal adequate constructions promote high electrical consumptions. It is for this reason that a lot of effort is made to reduce energy consumption in buildings, particularly in low-income dwellings.

Since 2002, the construction of dwellings for low-income families has been determined by the maximum cost of 117,063 times monthly minimum wage. It was meant to help the families with incomes of less than 3.9 times monthly minimum wage [3]. This type of dwelling represents an increase of the offer, which results in dwellings with a construction area of approximately 32 m², with a space for the living room, dining room and kitchen, and also one or two bedrooms and one bathroom. These dwellings are built with walls of common concrete block (0.12 m × 0.20 m × 0.40 m) and roofing "techos de vigueta y bovedilla" (mixed system with concrete joist and light polystyrene panel). The actual building system continues with the thermal inadequation problems, lack of thermal comfort, and high electric consumption. This seriously affects the economy of the families that live in those houses.

The Fig. 1 shows a prototype of low-income dwellings in Mexicali.

This article is looking forward to answer the next question: Is there any significant differences in the electric energy consumption that are caused by the designs of low-income dwellings?

The results presented in this paper are part of a research project called "Thermal Comfort and Energy Saving in Low-income Dwellings in Mexico: Hot Dry and Warm Humid Regions", which has been granted support from the National Housing Commission and National Council for Science and Technology.

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Fig. 1. Low-income dwellings, Mexicali, Mexico. Source: Corral, 2009.

2. Methodology

A field study was conducted in low-income dwellings in Mexicali city. The surveys were applied to identify the main characteristics of low-income dwellings, which included: physical conditions of the dwellings, the user profile and the thermal comfort perception. The characteristics of the dwellings were revised with the information given by the executive plans of the dwelling projects. The energy consumption registers were obtained through the Federal Electricity Commission. Then, with this information, the models of the actual low-income dwellings were identified [4].

The comfort perception of the occupants was done through the subjective judgments of the perceived thermal sensation respecting to the thermal environment condition from the subjects, it is worth mentioning that the appreciation was only from one of the dwelling users. The scale of thermal sensation values were: 1, not comfortable; 2, comfortable; 3, very comfortable. The measure method was based in ISO 10551:1995 [5].

3. Results

3.1. Electric energy consumption in low-income dwelling

The Fig. 2 shows of the monthly average electric energy consumption produced by the low-income dwelling in Mexicali. The highest electric energy consumption was observed during the months of July and August, 970 and 971 kWh, respectively. There is a differential of more than five times between the lowest and highest energy consumption.

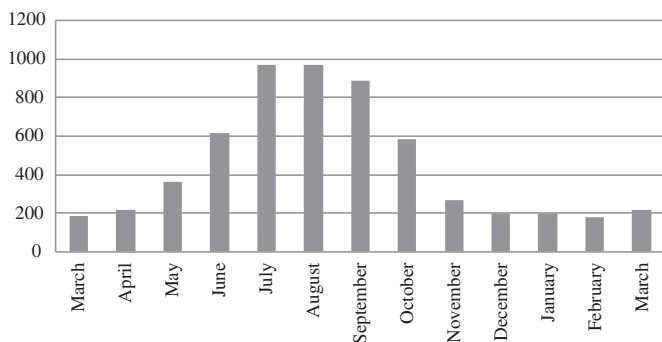


Fig. 2. Average monthly energy consumption, low-income dwelling, Mexicali (kWh).

Table 1

Average energy consumption by summer and winter, low-income dwellings, Mexicali.

Period	Average monthly energy consumption (kWh)
Summer (May–October)	730.67
Winter (November–April)	209.14
Differential	521.53

Source: Calculations based on data from the Federal Electricity Commission, 2009.

In Table 1, it is shown the average electric consumption between summer and winter periods and there is also a differential between both periods, which corresponds mainly to the electric consumption by the use of air conditioning (521.53 kWh).

3.2. Physic characteristics of low-income dwelling

In Mexicali, there were identified 21 prototypes for low-income dwellings. The percentage of each dwelling type is shown in Fig. 3, and in Fig. 4 the prototypes for low-income dwellings are shown.

3.3. Electric energy consumption and its impact in the dwelling, user and comfort perception in the low-income dwelling

3.3.1. Electric energy consumption per dwelling prototype

The electric energy consumption, the building area for the dwelling construction, amount of users, air-conditioning equipment and thermal comfort perception are shown in Table 2.

Based on the previous table, it is observed that even when the surfaces of the distinct types of dwellings are similar, the prototypes present significant differences in the annual energy consumption, as shown in Table 3.

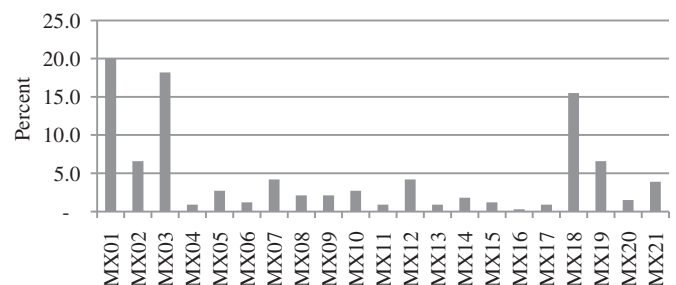


Fig. 3. Prototypes of low-income dwelling, Mexicali.

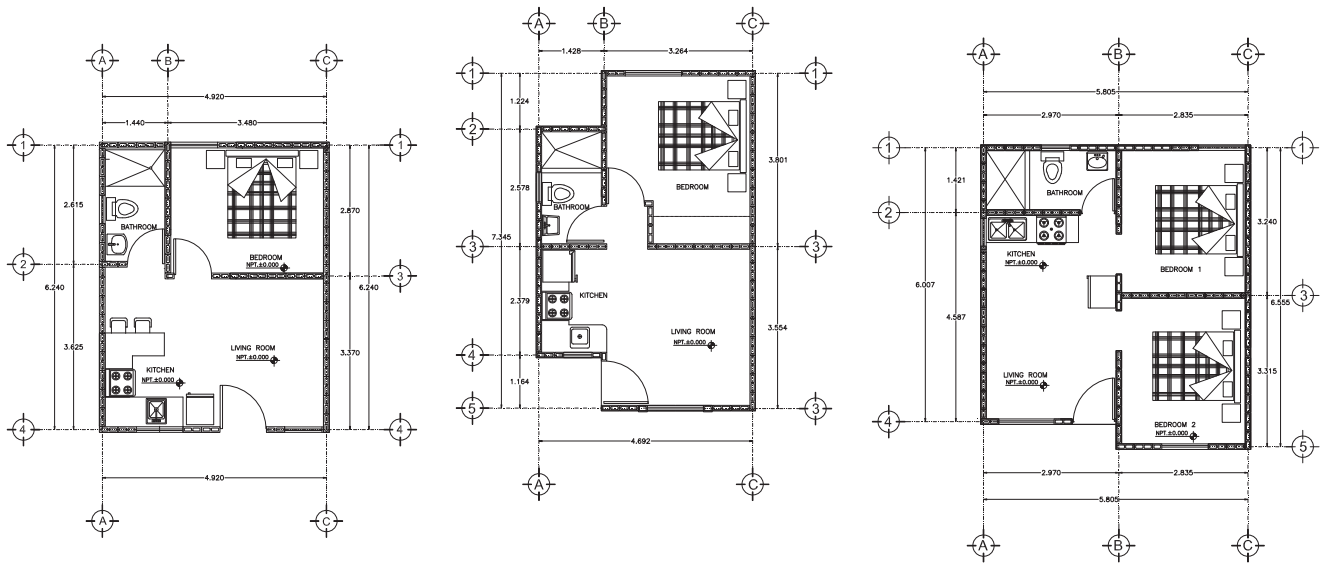


Fig. 4. Architectural plans prototypes low-income dwelling, Mexicali.

Table 2

Energy consumption, area, occupant, air-conditioning equipment and thermal comfort perception, according low-income's prototypes, Mexicali.

Prototype	Energy consumption (August) kWh	Annual electrical energy consumption kWh	Area	Occupant	Air-conditioning equipment	Summer thermal perception
MX 01	974	5394	32.5	3	W	1
MX 02	1047	6126	32.5	4	W	1
MX 03	982	5594	32.3	4	W	1
MX 04	835	4868	32.3	4	W	2
MX 05	1098	6605	34.2	4	W	1
MX 06	1017	7815	33.8	5	W	1
MX 07	1061	5460	32.4	4	W	1
MX 08	1087	6723	36.0	5	W	1
MX 09	1264	7052	34.7	3	W	1
MX 10	1118	6727	36.9	4, 5	W	1
MX 11	1062	6841	32.5	5	W	1
MX 12	909	5393	39.9	3, 5	W	1
MX 13	1266	7304	49.1	3, 4, 5	W + C	1
MX 14	964	5743	47.5	3, 4	W	1
MX 15	1005	5708	35.9	5	W	1
MX 16	1329	7931	35.9	6	W	1
MX 17	546	3203	43.4	3	W	1
MX 18	972	6017	34.5	4	W	1
MX 19	997	5847	33.5	4	W	1
MX 20	1436	8206	35.3	4, 5	W	1
MX 21	835	5468	34.6	3	W	1

Note: W, window-type air conditioners; W + C, windows-type air conditioners and cooler. 1, uncomfortable; 2, comfortable and 3, very comfortable.

There are differences of consumption that ascend to 3 times per square meter between some of the prototypes, for example: MX 020 consumes 232 kWh year/m² and MX 17 consumes 74 kWh year/m².

3.3.2. Thermal perception in low-income dwelling

The thermal perception in low-income dwellings during summer was mostly “uncomfortable” in 81.2% of the cases,

“comfortable” in 17.6% and “very comfortable” in 1.2%. According prototypes, the thermal perception is shown in Fig. 5.

Based on the previous, it is shown that the most critical or “less comfortable” prototypes, according to the user's thermal perception, were models MX 02, MX 06, MX 07, MX 09, MX 11, MX 16, MX 17 and MX 20. The rest of the prototypes are in “uncomfortable” conditions, but some considered them as “comfortable”. The only prototype in which it is manifested a complete comfort situation, was MX 04.

Table 3

Annual energy consumption index, low-income dwelling prototypes, Mexicali (kWh/m²).

High consumption		Low consumption		Common prototype (%)	
Prototype	kWh/m ²	Prototype	kWh/m ²	Prototype (%)	kWh/m ²
MX 09	203	MX 17	74	MX 01 (20%)	166
MX 11	210	MX 14	121	MX 03 (18.2)	173
MX 16	221	MX 12	135	MX 18 (15.6)	174
MX 06	231	MX 13	149	MX 02 (8.4)	173
MX 20	232	MX 04	151	MX 19 (6.6)	175

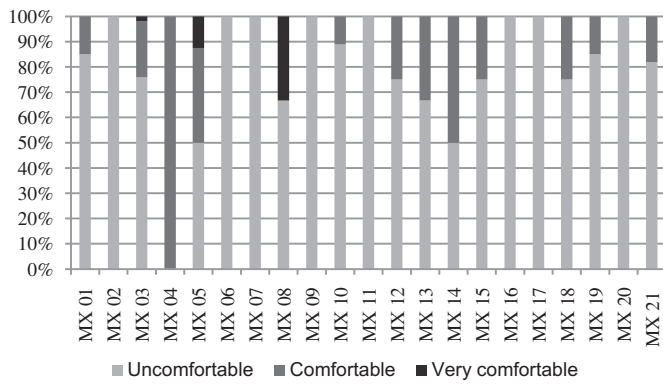


Fig. 5. Occupant's thermal perception, according low-income dwelling prototype, Mexicali.

4. Conclusions

The electric consumption of low-income dwellings shows a seasonal pattern similar to the ones presented by Mexicali

dwellings. The average consumption increases during summer almost four times more than it does during winter.

Average electric energy consumption is measured by a different square meter, even when the distinct prototypes have similar areas; this is because of the impact that comes from the user conduct.

Even when there is a variation between the seasonal consumption, it is shown that the thermal environment perception does not have a significant variation, which is due to the effect of thermal experience and expectation.

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